Paleomagnetic Correlation of Basic Intrusions in the Precambrian of Southern Africa

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Abstract. Samples of pre-Karroo dolerites have been collected from twenty sites in Bechuanaland and the Transvaal. After partial demagnetization in alternating magnetic fields, thirteen of the sites give directions of magnetization which form a close group. Some of the dolerites sampled have previously been classified as post-Ventersdorp or post-Transvaal. The measurements show, however, that the majority of the dolerites are post-Waterberg in age. The *post-Waterberg diabases* are therefore more widespread than had previously been supposed. The mean pole position for these diabases coincides almost exactly with that for the *Umkondo dolerites* in Rhodesia, and a statistical comparison of the paleomagnetic data confirms the difference between the pole positions is not significant. This agreement strongly supports contemporaneity of the igneous activity in these two regions, which are separated by some 1000 km. It is further suggested that the Umkondo and Waterberg systems should be correlated, which would imply that the 'post-Waterberg' igneous activity should be included within the Waterberg system. The age of the Umkondo and Waterberg systems is discussed in the light of the paleomagnetic data and recent age determinations. Paleomagnetic measurements on samples from the kimberlite pipe and dolerite sill in the Premier Mine, north of Pretoria, suggest that the age of this pipe is not Cretaceous, as has previously been supposed, but is also confined within the limits of the Waterberg system.

INTRODUCTION

The Rhodesian shield contains several isolated unfossiliferous Precambrian formations that have no direct connection with similar formations of the Transvaal shield in South Africa. Correlation between these two shield areas has in the past been largely based upon lithological similarities. It is well known that correlations based upon lithology are highly speculative, but in the absence of any other information this is the best that can be done.

A previous investigation of the Precambrian dolerites of Rhodesia by *McElhinny and Opdyke* [1964] has shown that these intrusions represent two distinct periods of magmatic activity. The paleomagnetic directions of magnetization formed two widely separated groups, which indicated the possibility of using the paleomagnetic method in its correlative aspect in southern Africa.

Figure 1 shows a simplified geologic map of southern Africa. The Rhodesian Precambrian dolerites consist of the *Mashonaland dolerites* and the *Umkondo dolerites*, as distinguished by *McElhinny and Opdyke* [1964]. The former intrude the Great Dyke and the granites and gneisses of the Rhodesian shield; the latter are associated with the Umkondo system on the eastern border of Rhodesia. To the southwest, in the Bechuanaland Protectorate, large numbers of dolerite sills and dikes intrude various Precambrian formations correlated with the Waterberg or Transvaal systems in South Africa. Others intrude various Precambrian granites, including the Gaberones granite batholith on the eastern border with South Africa. In South Africa the most widespread of the pre-Karroo dolerites intrude the Waterberg system (*post-Waterberg diabase*) in the Transvaal. Other dolerites intrude the Transvaal and Witwatersrand systems, and these in the past have been classified as post-Transvaal or post-Witwatersrand.

McDougall [1963] determined some K-Ar ages of Karroo dolerites in South Africa. Two of his determinations from South Africa and one from the Bechuanaland Protectorate were shown to be of Precambrian rather than Karroo age, and an age of around 1000 m.y. was indicated from his measurements (see Table 1). The possibility therefore existed that there was a correlation between either of the Rhodesian

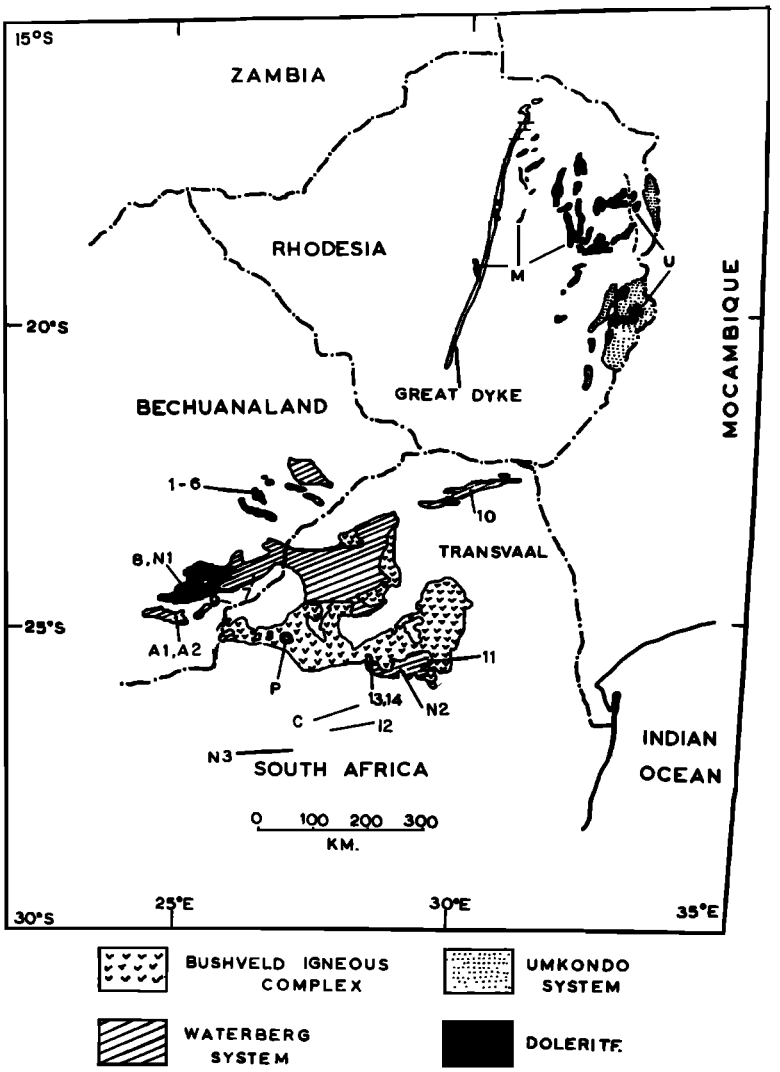


Fig. 1. Geological sketch map of southern Africa showing sampling sites. *M*, Mashonaland dolerites; *U*, Umkondo dolerites; *P*, Pilansberg alkaline complex.

Precambrian dolerites and those in South Africa and Bechuanaland. A distance of some 1000 km separates these two areas, and a successful correlation over this distance could open the possibility of using the paleomagnetic method to correlate the major Precambrian formations in Rhodesia and South Africa. Alternatively, the ages of these dolerites might differ from the ages of those in Rhodesia, and paleomagnetic measurements would then provide a new point on the African Precambrian polar-wander curve. For this reason many samples of dolerites from Bechuanaland and South Africa were collected during February 1964.

Sampling

Samples were collected from twelve sites in the Bechuanaland Protectorate, seven in the Transvaal, and one in the Orange Free State. These are shown in Figure 1, and full sampling details are given in Table 1 together with the previously accepted geologic ages of the sampled bodies.

Three methods of sampling were used. At

TABLE 1. Sampling Details

Site No.	Site Name	Site Position Lat. S	Site Position Long. E	Sampling Method	Sampling Number	Body	Exposure	Intrusive into	Geologic Age of Body	Reference
1	Shushong sill	23°07′	26°36′	L	4	Sill, 50 km long and 150 m in exposed thickness	Surface outcrop	Shushong series	Post-Transvaal Bushveld complex?	*Cullen* [1961]
2	Shushong sill	23°03′	26°33′	L	4		Surface outcrop			
3	Shushong sill	23°00′	26°25′	L	4		Surface outcrop			
4	Shushong sill	23°00′	26°25′	S	6		Stream section			
5	Shushong sill	23°00′	26°25′	L	4		Surface outcrop			*Grubb* [1961]
6	Shushong sill	22°54′	26°24′	L	4		Surface outcrop			
7	Molotoana River	24°20′	26°08′	L	4	Sill	Stream section	Waterberg system	Post-Waterberg	
8	Linchwe le Tau	24°14′	25°52′	L	4	Sill	Surface outcrop	Waterberg system	Post-Waterberg	
9	Khale sheet	24°40′	25°53′	S	6	Sheet, 50 m thick	Quarry	Gaberones granite	Karroo Post-Waterberg, Pre-Karroo 900 m.y.	*Eales* [1959] *Boocock* [1959] *McDougall* [1963]
10	Wyllie's Poort	22°55′	29°56′	B	5	Sill	Road cutting	Waterberg system	Post-Waterberg	Sheet 42. 1:250,000 *G.S.S. Africa*, 1956
11	Mooiplaats 242-JS	25°30′	29°28′	S	6	Sill	Stream section	Waterberg system	Post-Waterberg	Sheet 2529C. 1:125,000 *G.S.S. Africa*, 1962
12	Barnard's Kop	26°54′	28°41′	S	6	Sill	Surface outcrop	Ventersdorp system	Post-Ventersdorp 1050 m.y.	*McDougall* [1963]
13	Premier Mine	25°42′	28°32′	S*	10	Sill	Underground in mine	Kimberlite pipe	Post-Waterberg	
14	Premier Mine kimberlite	25°42′	28°32′	B	2	Kimberlite pipe	Underground in mine	Altered Transvaal rocks	Post-Waterberg	Sheet 2528D. 1:125,000 *G.S.S. Africa*, 1962
N1	Molepolole	24°23′	25°33′	L	4	Sill	Surface outcrop	Waterberg system	Post-Waterberg	
N2	Doornrug 302-JS	25°53′	29°01′	S	5	Sheet	Stream section	Waterberg system	Post-Waterberg	Sheet 2529C. 1:125,000 *G.S.S. Africa*, 1962
N3	Annas Rust	26°52′	27°30′	S	6	Sill	Stream section	Archaean granite	Post-Vredefort dome	*Poldervaart* [1962]
A1	Moshaneng Mine	24°56′	25°17′	B	4	Sill	Underground in mine	Transvaal system	Post-Transvaal	
A2	Kanye	24°58′	25°20′	S	7	Sill	Stream section	Dominion Reef system	Post-Transvaal	
C	Van Dyk mine	26°13′	28°15′	B	8	Dike	Mine, 1400 m underground	Witwatersrand system	Post-Witwatersrand, 1120 m.y.	*McDougall* [1963]

four sites, unweathered blocks were collected and oriented with a compass (method B in Table 1). From eight sites, cores 60 cm long and 2.5 cm in diameter were obtained (method L in Table 1), and from seven sites, cores 15 cm long and 2.5 cm in diameter were obtained (method S in Table 1). At the latter fifteen sites the portable drilling equipment described in detail by *McElhinny and Neal* [1966] was used, and the cores were oriented with respect to the sun's azimuth using the method described by *Gough and Opdyke* [1963].

In the Premier Diamond Mine at Cullinan, northeast of Pretoria, an 80-m-thick dolerite sill cuts across the kimberlite pipe in which mining operations take place. The pipe contains a large xenolith which is generally accepted by the Geological Survey of South Africa to be of Waterberg quartzite. The mine management kindly arranged to drill ten cores underground at three positions using the mine drilling equipment. The drills were so oriented that all cores were horizontal and due north. The cores were each 2.2 cm in diameter and about 8 cm long (method S* in Table 1). Four cores were drilled from the center of the sill and four near the contact with the kimberlite. A further two were drilled in the baked kimberlite about 3 m from the base of the sill. The two blocks from site 14 were collected from the unbaked kimberlite 50 m above the top of the sill.

It was not possible to obtain samples of the dike in the Van Dyk Consolidated Gold Mine at Brakpan at exactly the same position as *McDougall* [1963] did (16 level west). Instead, the same dike was sampled at 22 level west, where a drive cuts through it. The dike has a width of about 30 m at this position and the samples were collected across its whole width. Because the magnetization of the dike itself might have caused an appreciable anomaly in the magnetic declination, thereby causing errors in the orientation of the blocks, the mine survey very kindly made a detailed series of 28 measurements of the declination through the dike at 22 level west. The measurements gave a mean value of 14¼° west of north for the declination, compared with 17° west of north at the surface.

Measurements

From the oriented blocks two cores, 2.5 cm in diameter, were drilled in the laboratory. From each of these cores one cylinder, 2.5 cm in length, was cut. For the purpose of the statistical analyses which follow, the mean of these two specimens was regarded as a single observation (sample). From the bottom of a 15-cm core, one cylinder, 2.5 cm long, was cut, and this was regarded as one sample for the analysis. From a 60-cm-long core, two cylinders, 2.5 cm in length, were cut, one from the bottom and the other at the 30-cm mark. It is believed that the distance of 30 cm separating these two cylinders justifies regarding them as two separate samples for the analysis. The number of samples at each site as defined above is indicated in Table 2, which gives the results of the measurements described below.

The direction and intensity of the natural remanent magnetization (NRM) of each of the cylinders was measured with a spinner magnetometer [*Gough,* 1964]. The specimens from all the sites were subjected to alternating magnetic field demagnetization using the improved apparatus recently described by *McElhinny* [1966*a*] and following the criteria for magnetic cleaning outlined by *McElhinny and Gough* [1963].

At three sites, N1, N2, and N3 (Table 1), the directions of magnetization of the samples were initially widely scattered, and the application of alternating magnetic fields produced no improvement in precision. It is possible that these sites have been completely remagnetized by lightning effects, a not uncommon phenomenon in southern Africa.

At all the others, the secondary components of magnetization were successfully removed in fields ranging from 10 to 60 milliwebers m^{-2} peak value (see Table 2). As has previously been observed, only low fields were required when the secondary component was of the viscous type, whereas higher fields were necessary to remove isothermal components due to lightning currents. Two samples (from the same 60-cm core) from site 7 were rejected. These did not respond to alternating field demagnetization, and the directions of magnetization differed widely from those of the other six samples from the same site. It seems probable that the core was struck by lightning. The intensity recorded for these two samples was much greater than those of the other six samples from the same site.

At each of the three sites 9, A2, and C, the

NRM directions formed a close group. After partial demagnetization each of these groups moved as a whole away from the direction of the present earth's field and each became more tightly grouped. Each sample has apparently acquired a viscous component of magnetization. The component added represents almost the same fraction of the original magnetization in each case, with the result that the directions of magnetization still form a close group. Thus even good groupings of the NRM of samples from the same site cannot be regarded as evidence of their stability. This serves to emphasize the importance that must be attached to partial demagnetization techniques as a means of testing the stability of the NRM directions and of removing secondary components of magnetization. The mean direction of magnetization after magnetic cleaning at site 12 differs by almost 180° from those of sites 1 to 11, 13 and 14.

Interpretation of Results

Post-Waterberg diabases. The site mean directions of magnetization after magnetic cleaning are illustrated in the stereonet in Figure 2, which shows that all the sites numbered 1 to 13 (Figure 1, Tables 1 and 2) form a close group, with directions of magnetization nearly due south and having shallow dips. In the vicinity of site 10, the Waterberg red beds that have been intruded by the dolerite sill, dip at 24° toward the north (5° east of north). In this region the similar folding of the overlying Karroo sediments proves that the age of the folding was post-Karroo. The data presented in Table 2 and Figures 2 and 3 relating to this site have therefore been corrected for the attitude of the adjacent sediments. At the other sites, the intruded beds were either flat lying or dipped at not more than 10° in the east-west direction. Since the directions of magnetization are nearly horizontal in a north-south direction, the correction for the attitude of the adjacent sediments makes no significant difference to the directions of magnetizations. Sites 1 to 13 include all the post-Waterberg diabases from which results were obtained and some diabases which have been classified as post-Transvaal or post-Ventersdorp.

The stability of the magnetization of these

TABLE 2. Site Directions for the Precambrian Basic Intrusives of Bechuanaland and the Northern Transvaal

Site No.	N	M_N D	I	R	k	B_{ac}, mwb m^{-2}	M_{ac} D	I	R	k	α_{95}	Site Pole Lat.	Long.	N or S
1	8	177½	+5½	5.736	3.1	30–50	191½	+14	7.961	180	4.2	70.7°N	62 7°E	S
2	7	30	+34½	4.390	2.3	40	190½	+9½	6 991	674	2.3	69.1°N	56.6°E	S
3	8			3.584*		30–50	190½	+4	7.991	796	2 0	66.6°N	53.7°E	S
4	6			3 228*		50	187	+5	5.980	254	6.7	68.5°N	46.1°E	S
5	7			2.536*		50	187	+5	6.981	323	3.3	68.4°N	45.7°E	S
6	8	326	+31	5.186	2.5	30	186½	+½	7.992	897	1.8	66.4°N	42.8°E	S
7	6	70	−29	5.685	3 0	30–50	193½	−5½	5.970	165	5.2	59 9°N	54.2°E	S
8	7	150	−53½	4.994	3 0	50–60	191	−33	6.545	13.2	17	46 5°N	41.3°E	S
9	6	193	−38½	5.942	86	15	187½	−13	5.987	379	3.4	57.7°N	40.3°E	S
10	5	218	−37	4.747	16	20	194	+24	4.940	66 5	9.5	73 1°N	84 2°E	S
11	6			3.118*		30	188	+6	5.633	13.6	19	66.2°N	50.2°E	S
12	6	85½	−3	4.071	2 6	20–50	16	−14½	5.983	292	3.9	65.3°N	69 3°E	N
13	10	190½	−27	8 565	6.3	10–30	183	−3	9.878	73.5	5.7	62.7°N	34 6°E	S
14	2	202	−23	1.987	80	20	204½	−18	1.987	78.1	29	47.7°N	65 8°E	S
A1	4	333	−7½	3 668	9.0	50	329½	+8½	3.988	244	5.9	48.2°E	155.5°E	S
A2	7	148½	−56½	6 874	48	30	139½	−39	6.979	284	3.7	28.7°N	18.0°W	S
C	8	144	−71	7 884	60	25	158½	−66½	7.950	140	4.7	12 4°N	13.9°E	S

N, number of samples measured.
R, length of the resultant of the N unit vectors.
D, declination of remanent magnetization.
I, inclination of remanent magnetization.
M_N, NRM.
M_{ac}, RM after demagnetization in alternating field B_{ac}.
$k = (N - 1)/(N - R)$ = estimate of Fisher's κ parameter.
α_{95}, semiangle of cone of 95% confidence for mean direction.
* These numbers refer to the 'resultants' of data which are random on *Watson's* [1956b] criterion.

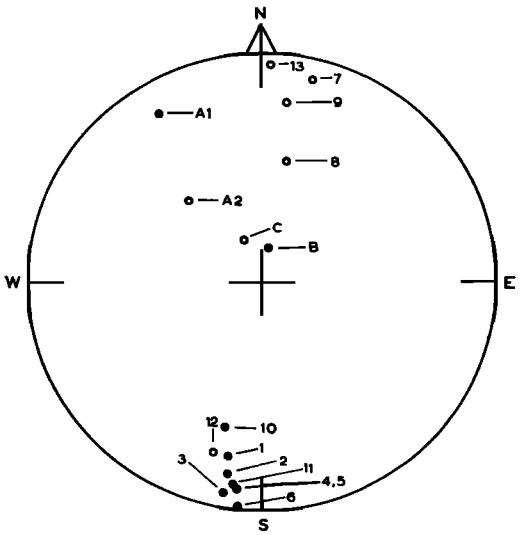


Fig. 2. Mean directions of magnetization at sixteen sites in the Precambrian dolerites from Bechuanaland and South Africa. Lower-hemisphere stereographic projection; solid circles, north-seeking directions down; open circles, south-seeking directions down.

rocks has been well tested by alternating field demagnetization. The magnetization therefore is probably the TRM acquired when the rocks originally cooled. The measurements therefore indicate that all these diabases are of the same age and that the post-Waterberg igneous activity must have been fairly widespread, including all the major exposures of Precambrian dolerites in Bechuanaland and the Transvaal. These intrusives cover an area of 750,000 km^2.

The parameters of the mean direction of magnetization of these sites are: $N = 13$ sites, $D = 189.6°$, $I = +2.3°$, and $R = 12.616$, where the symbols are the same as for Table 2. Site 12 has a direction of magnetization almost exactly reversed with respect to the other sites, and, in calculating the mean, the direction of magnetization at this site was changed by 180°.

The estimate k of *Fisher*'s [1953] precision parameter κ for the site directions is 31.3, and the corresponding angular dispersion Δ, given by $\Delta = 81/k^{1/2}$, is 14.5°. The use of an F test gives upper and lower limits k_1 and k_2 such that there is 95% confidence that $k_1 \geq \kappa \geq k_2$ [*Gough et al.*, 1964]. This provides an upper limit at the 95% confidence level for the value of Δ, namely 20.5°. This is greater than that calculated by *Creer* [1962] for the present earth's field at equatorial latitudes (19.0°). The estimate of the angular dispersion must therefore be considered not significantly different from that calculated for the present earth's field.

Comparison of the post-Waterberg and Umkondo intrusives. It will be noted in Table 3 that the mean pole positions for these two sets of intrusions are very close to one another. Figure 3 shows a polar stereographic projection of the thirteen *site pole positions* for the post-Waterberg diabases. On the same stereonet are plotted circles of 95% confidence for the Umkondo dolerites [*McElhinny and Opdyke*, 1964] and for the combined Umkondo lavas and dolerites [*McElhinny*, 1966*b*]. The agreement between these sets of data is remarkable, and it suggests that the Umkondo and Waterberg igneous activities were contemporaneous.

The agreement between the poles for the Umkondo and post-Waterberg intrusives can be tested statistically by the method given by *Watson* [1956*a*] and *Watson and Irving* [1957]. In this test, however, it is assumed that the dispersions of the two populations are the same. It is thus necessary first to test for identity of dispersions.

The $N_1 = 13$ sites of the post-Waterberg intrusives give an estimate $K_1 = 87.0$ for the precision of the site poles, whereas the $N_2 = 10$ sites of the Umkondo intrusive give an estimate $K_2 = 66.0$ for the precision of these site poles. The appropriate equation is

$$F_{2(N_2-1),2(N_1-1)} = K_1/K_2$$

giving, in the present case,

$$F_{18,24} = 1.32$$

which is not significant. Thus the data give no reason for supposing the two populations have different dispersions.

To test the identity of the two polar vectors it is supposed that the $N_1 = 13$ site poles have resultant of length R_1 and the $N_2 = 10$ site poles have resultant of length R_2. The resultant of all the $N = 23$ site poles has length R. The statistic

$$(N - 2)(R_1 + R_2 - R)/(N - R_1 - R_2)$$

may be referred to the F-ratio tables with 2 and

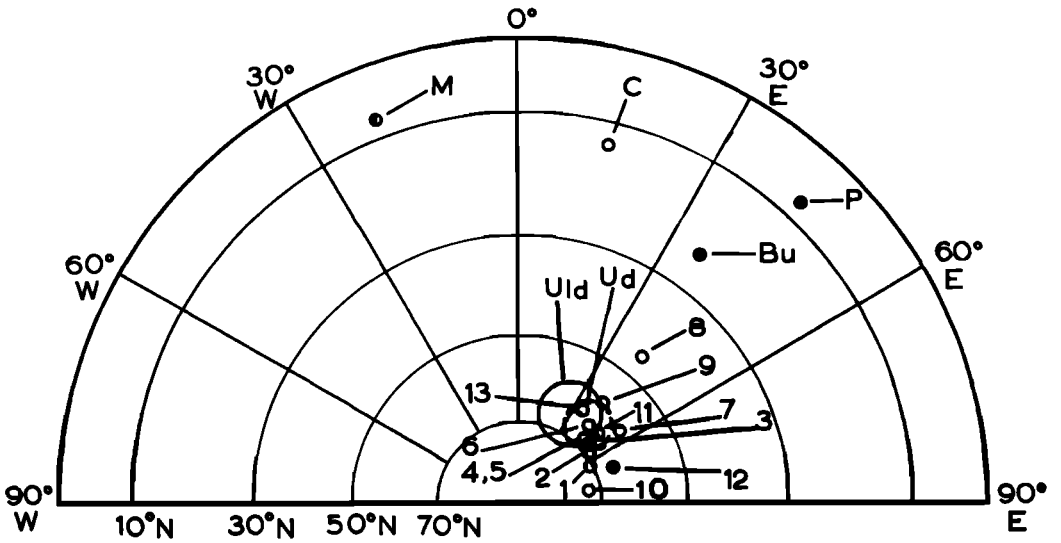


Fig. 3. Pole positions for fourteen sites in the Precambrian dolerites from Bechuanaland and South Africa. A polar stereographic projection is used, and the mean pole positions for the Mashonaland dolerites (*M*), the Pilansberg dikes (*P*), and the Bushveld gabbro (*Bu*) are also indicated. Solid circles, north poles; open circles, south poles; *M* has mixed polarity. Solid-line circle (*Uld*) shows 95% confidence for the mean pole of the combined Umkondo lavas and dolerites. Dashed-line circle (*Ud*) shows 95% confidence for the mean pole of the Umkondo dolerites only.

$2(N - 2)$ degrees of freedom, giving, in this case,

$$F_{2,42} = 0.80$$

which is not significant. Thus the data give no reason for supposing the polar vectors are different.

This similarity of the Umkondo dolerite and post-Waterberg diabase pole positions strongly supports contemporaneity within a period of the order of 10^7 years. It appears, therefore, that the igneous activity at this time of the Precambrian covered a very large part of southern Africa and included a region from 17 to 27°S and from 25 to 33°E.

Correlation of the Umkondo and Waterberg systems. The Umkondo system is a shallow-water marine deposit [*Swift*, 1962; *Watson*, 1964] consisting mainly of limestones at the base, followed by quartzites and basalt lava flows. The lavas represent the uppermost sequence of the system.

McElhinny [1966*b*] has shown that the Umkondo lavas and dolerites have similar directions of magnetization, which supports the geological view that they are of the same age. Since the paleomagnetic data presented in this paper suggest that the Umkondo and Waterberg intrusives were contemporaneous, it is tempting to correlate the Umkondo and Waterberg systems themselves. The Waterberg sediments (which consist largely of terrigenous material, including red beds) could then be interpreted as the continental facies of the marine Umkondo to the north. The Umkondo lavas and dolerites have already been correlated with the post-Waterberg diabases, and it would thus seem more logical to include these diabases *within* the Waterberg system. Support for this hypothesis may be obtained by a paleomagnetic study of the Waterberg red beds. Such an investigation is planned.

The Premier Mine sill and the kimberlite pipe. Figure 4 shows a stereographic projection of the directions of magnetization of the samples collected underground in the Premier Mine at Cullinan. The eight samples from the dolerite sill intruding the kimberlite pipe, and the two from the baked contact in the kimberlite, form a good group of directions similar to those of the other post-Waterberg diabases sampled elsewhere. The dolerite sill clearly belongs to the

TABLE 3. Pole Positions Calculated by Combining Site Pole Positions

	N	R	K	A_{95}, deg	Lat., deg N	Long., deg E
Post-Waterberg diabases	13	12.862	87	4½	65	50½
Umkondo dolerites	10	9.864	66	6	65	42½

N is the number of sites; R is as in Table 2; K is the precision of site poles; A_{95} is the semiangle of the cone of 95% confidence for the mean pole position.

same period of igneous activity, and this site (13) has already been included in the over-all calculations made above.

The kimberlite pipe contains a xenolith of Waterberg quartzite and is therefore younger than the base of the Waterberg system and older than the sill. It has already been suggested that the sill should be included within the Waterberg system, so that the age of the kimberlite is also apparently confined within the limits of the Waterberg system. This appears to be the first evidence that the kimberlite pipes of southern Africa are not all of Cretaceous age as has previously been supposed [e.g., *Holmes*, 1936; *Dawson*, 1962].

The two samples from the unbaked kimberlite (site 14) have a mean direction of magnetization which differs from that of the sill and its baked contact by 26°. A comparison of these two polar vectors by the method given by *Watson* [1956*a*] shows that there is less than 0.1% probability that they are drawn from the same population. The difference between the two directions of magnetization is therefore highly significant, and the unbaked kimberlite and the sill must have been magnetized at different times. This, of course, is not surprising; the kimberlite obviously predates the sill. This significant difference does not necessarily imply a large age difference, and this point might be elucidated by further sampling of the pipe.

Other intrusives. Sites A1 and A2 were sampled from the same sill in the area west of Kanye in Bechuanaland. Both sites exhibit stable directions of magnetization which differ widely from one another (Figure 2). Neither of the observed directions agrees with any of the other available data (Figure 2), and the results are somewhat puzzling. It is possible that there has been relative movement between different parts of the sill, and this point is to be investigated by further sampling.

The dike in the Van Dyk Mine (site C) varies from being doleritic at the margins to syenitic at the center, a feature which has also been noted in the Pilansberg dikes [*Gough*, 1956]. It has a direction of magnetization which cannot be correlated with any of the other sites sampled (Figure 2), and its pole position lies almost halfway between those of the Mashonaland dolerite and the Pilansberg dikes [*Gough*, 1956] (Figure 3). It is therefore not possible to draw any conclusions from the paleomagnetic measurements made on samples from this dike.

The Age of the Umkondo and Waterberg Systems

The ages determined by the K-Ar method for a number of samples are given in Table 4. In

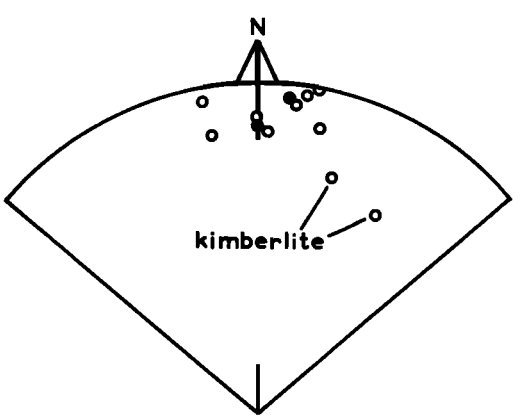


Fig. 4. Stereographic projection of the directions of magnetization of all samples at sites 13 and 14 in the Premier Mine. The two samples of unbaked kimberlite (site 14) are indicated. Open circles, south-seeking directions on lower hemisphere; solid circles, south-seeking directions on upper hemisphere.

TABLE 4. Potassium-Argon Ages of Some Precambrian Dolerites from Southern Africa

	Paleomagnetic Site	Sample Analyzed	Age, m.y.	Source
Umkondo dolerites	B*	Pyroxene	715±70	*Snelling et al.* [1964]
The site designations	E	Whole rock	640±20	Miller (private communication)
are those of *McElhinny*	G	Whole rock	1136±9	Miller (private communication)
and *Opdyke* [1964].	H	Whole rock	757±20	Miller (private communication)
Bechuanaland and Trans-	2	Plagioclase-pyroxene	590±24	Snelling (private communication)
vaal dolerites	6	Plagioclase-pyroxene	660±26	Snelling (private communication)
These are all suggested	7	Plagioclase-pyroxene	940±38	Snelling (private communication)
by the authors as being	9	Plagioclase	900±36	*McDougall* [1963]
post-Waterberg and	11	Plagioclase-pyroxene	870±35	Snelling (private communication)
equivalent to Umkondo.	12*	Whole rock	1050±42	*McDougal* [1963]
	N1	Plagioclase-pyroxene	1255±50	Snelling (private communication)
	N2	Plagioclase-pyroxene	865±35	Snelling (private communication)
Other dolerites	A1	Plagioclase-pyroxene	1100±44	Snelling (private communication)
	C*	Feldspar	1120±45	*McDougall* [1963]

* The samples for paleomagnetic measurements were taken from the same bodies as were sampled for age determinations but at slightly different locations. The errors quoted are experimental errors.

spite of the good agreement of the paleomagnetic results, the age determinations range from 590 to 1255 m.y. If the paleomagnetic correlation is accepted as valid, and there appears no reason at present to doubt it, it seems that the samples on which the K-Ar measurements were made have lost a considerable amount of argon.

The Waterberg system is younger than the Bushveld igneous complex, which has been well dated by *Nicolaysen et al.* [1958] at 1950 ± 50 m.y. This therefore sets a maximum age for the Waterberg and Umkondo igneous activity. The minimum age of the Waterberg system is given by the age of the Leeuwfontein alkaline complex, which intrudes the Waterberg sediments and has been given an age of 1420 ± 70 m.y. by the lead isotope method [*Oosthuyzen,* 1964]. Further, *McElhinny and Opdyke* [1964] suggest that the Umkondo dolerites are older than the Mashonaland dolerites, which have a minimum K-Ar age of 1638 ± 26 m.y. This was based on the evidence at one site, where the Mashonaland dolerites were found apparently intruding the Umkondo dolerites, and it would therefore set a minimum age to the Umkondo and Waterberg igneous activity at 1612 m.y. Doubts have been expressed, however, in various personal communications, as to the validity of the observed age relationship between the Umkondo and Mashonaland dolerites.

Oosthuyzen [1964] has made an age determination of 1790 ± 70 m.y., using the lead isotope method, on a sample of granophyre that intrudes sediments which are correlated either with the Waterberg or Loskop systems. If the sediments can be correlated with the Waterberg system, the age limits of this system and the Umkondo system lie between 1720 and 2000 m.y., which is still consistent with the evidence presented to date. Paleomagnetic measurements on the granophyre sampled by Oosthuysen might help in differentiating between these possibilities. These are to be attempted shortly.

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